

Tensor Products

Exercise 1. Less talked about than inner products is the **outer product** of two vectors. If $u, v \in \mathbb{R}^n$, their outer product is $\langle u, v \rangle = uv^T \in M_n(\mathbb{R})$.

- (a) Show that the map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow M_n(\mathbb{R})$ given by the outer product is bilinear. That is,
 - i. $\langle u_1 + u_2, v \rangle = \langle u_1, v \rangle + \langle u_2, v \rangle$, and similarly for addition in the second component; and,
 - ii. for any scalar a , $\langle au, v \rangle = \langle u, av \rangle = a\langle u, v \rangle$.
- (b) Suppose V is a vector space, and suppose $\phi: \mathbb{R}^n \times \mathbb{R}^n \rightarrow V$ is any bilinear map. Show that there is a linear map $\tilde{\phi}: M_n(\mathbb{R}) \rightarrow V$ so that $\phi(u, v) = \tilde{\phi}(\langle u, v \rangle)$.
- (c) The usual inner product is a bilinear map $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. What is the corresponding map $M_n(\mathbb{R}) \rightarrow \mathbb{R}$?

Solution 1.

- (a) The first property is true because matrix multiplication distributes over matrix addition (and so does taking the transpose.) The second property is true because matrix multiplication is associative and scalars commute with matrices.
- (b) Let $e_1, \dots, e_n$ be the standard basis for $\mathbb{R}^n$. Notice that $\langle e_i, e_j \rangle$ is the matrix that is zero everywhere except for a single 1 at the (i, j) th position. In particular, $\langle e_i, e_j \rangle$ form a basis for $M_n(\mathbb{R})$. To define a linear map on $M_n(\mathbb{R})$, it suffices to define the map this basis, so set $\tilde{\phi}(\langle e_1, e_2 \rangle)$ to be $\phi(e_1, e_2)$. The bilinearity of ϕ guarantees that $\tilde{\phi}$ has the correct value on all outer products.
- (c) $\tilde{\phi}(A) = \text{Tr } A$.

Exercise 2. Let R be an integral domain with fraction field K , and let M be a free R module of rank n . Prove that any list of $n + 1$ elements of M are linearly dependent.

Solution 2. Let $m_0, \dots, m_n$ be a list in M . Let $V = K \otimes_A M$, and let $v_i = 1 \otimes_A m_i$. Note that since tensor products distribute over direct sums, V has dimension n , so there is some linear dependence

$$c_0 v_0 + \dots + c_n v_n = 0.$$

In fact, each c_i can be chosen to be in A rather than K , since we can multiply both sides by any denominators. But then

$$1 \otimes (c_0 m_0 + \dots + c_n m_n) = 0.$$

Since M is free, this implies that $c_0 m_0 + \dots + c_n m_n = 0$ as required.

Exercise 3. Prove that $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/3\mathbb{Z})$ is the trivial group. What about $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/4\mathbb{Z})$?

Solution 3. Let $x \otimes y$ be a simple tensor in $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/3\mathbb{Z})$. Notice that $3 \equiv 1 \pmod{2}$ so that $x = 3x$. Then

$$x \otimes y = (3x) \otimes y = x \otimes (3y) = 0.$$

We have no such luck trying to prove an arbitrary element of $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/4\mathbb{Z})$ is zero. Here's one way to see that this group is not trivial. There is a bilinear map

$$b : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$$

defined by

$$b(x, y) = xy \pmod{2}.$$

This bilinear map is not trivial; for example, $b(1, 1) = 1$. So by the universal property of tensor products, $(\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/4\mathbb{Z})$ cannot be trivial.

Exercise 4. Let A be a commutative ring, and let B be an A -algebra (that is, there is a ring homomorphism $\phi : A \rightarrow B$ which induces on B the structure of an A -module.)

- (a) Show that $B \otimes_A A[t] \cong B[t]$.
- (b) Recall that if I is an ideal of A , then the extension of I to B , denoted I^e , is the ideal of B generated by the image $\phi(I)$. Prove that $B/I^e \cong B \otimes_A (A/I)$.
- (c) Prove that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathbb{C} \times \mathbb{C}$ (Hint: rewrite the second $\mathbb{C}$ in the tensor product as $\mathbb{R}[x]/(x^2 + 1)$.)

Solution 4.

- (a) Note that as modules, $A[t]$ is a direct sum of countably many copies of A and $B[t]$ is a direct sum of countably many copies of B . So, since tensor products distribute over direct sums, these objects are at least isomorphic as B -modules. The isomorphism between them can be written as $b \otimes f \mapsto b\tilde{\phi}(f)$, where $\tilde{\phi}$ is the unique extension of ϕ to a map $A[t] \rightarrow B[t]$. This is an algebra homomorphism by, for example, the universal property. So the isomorphism is in fact an isomorphism of algebras.
- (b) The fastest way to do this is to use the exactness of the tensor product. The exact sequence

$$0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$$

becomes

$$B \otimes_A I \rightarrow B \rightarrow B \otimes_A (A/I) \rightarrow 0$$

when tensored with B . The image of $B \otimes_A I$ in B is exactly I^e , so that we have the short exact sequence

$$0 \rightarrow I^e \rightarrow B \rightarrow B \otimes_A (A/I) \rightarrow 0.$$

Exercise 5.

- (a) Let $a, b \in \mathbb{Z}$. There is a map $\mathbb{Z} \rightarrow \mathbb{Z}/a\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{b}]$ which sends 1 to $1 \otimes \frac{1}{1}$. Find the kernel of this map in terms of a and b .
- (b) Now let $p \in \mathbb{Z}$ be prime, and a be as before. As above, there is a map $\mathbb{Z} \rightarrow \mathbb{Z}/a\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}$. Find the kernel of this map in terms of a and p . (Recall: for a prime ideal $\mathfrak{p} \subset R$, $R_{\mathfrak{p}}$ means we put in all denominators which are *not in* $\mathfrak{p}$: $S = R \setminus \mathfrak{p}$.)

Solution 5.

- (a)

Exercise 6. (August 2020 Problem 4) Let R be a commutative ring.

- (a) Give an example of a free R -module M and an injective homomorphism $M \rightarrow M$ which is not an isomorphism. (For this part, R can be whatever commutative ring you like.)
- (b) If M_1 and M_2 are free of finite ranks r_1 and r_2 , show that a surjective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \geq r_2$.
- (c) Assume furthermore that R is an integral domain. Show that an injective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \leq r_2$.

Solution 6.

- (a) Recall that if M is finitely generated as an R -module, then any surjective endomorphism is injective. This problem shows that the converse is not true. For example, consider $\mathbb{Z}$ as a module over itself. Then the map $\mathbb{Z} \rightarrow \mathbb{Z}$ given by multiplication by 2 is injective but not surjective.
- (b) We have $M_1 \cong R^{r_1}$ and $M_2 \cong R^{r_2}$. Consider a map $\varphi : R^{r_1} \rightarrow R^{r_2}$, and let $\mathfrak{m}$ be a maximal ideal in R so that $R/\mathfrak{m} \cong k$ where k is a field. It is a fact that φ is surjective iff $\varphi_{\mathfrak{m}}$ is surjective for every maximal ideal $\mathfrak{m}$, so we may assume that R is a local ring with unique maximal ideal $\mathfrak{m}$. Then consider the commutative diagram

$$\begin{array}{ccc} R^{r_1} & \xrightarrow{\varphi} & R^{r_2} \\ \downarrow & & \downarrow \\ k^{r_1} & \xrightarrow{\bar{\varphi}} & k^{r_2} \end{array}$$

where the vertical maps are the natural quotient maps and $\bar{\varphi}$ is induced by φ . By Nakayama's Lemma, we know that $g_1, \dots, g_{r_2}$ are generators for M_2 iff $\bar{g}_1, \dots, \bar{g}_{r_2}$ are generators for k^{r_2} . Then φ is surjective iff there exists $f_i \in R^{r_1}$ such that $\varphi(f_i) = g_i$ for each $i = 1, \dots, r_2$. And this occurs iff there exists $\bar{f}_i \in k^{r_1}$ such that $\bar{\varphi}(\bar{f}_i) = \bar{g}_i$ for each $i = 1, \dots, r_2$. Since $\bar{\varphi}$ is a map of vector spaces, this happens iff $r_1 \geq r_2$.

- (c) Since R is a domain, (0) is a prime ideal, so let $K = \text{Frac}(R) = R_{(0)}$. Any map $\varphi : R^{r_1} \rightarrow R^{r_2}$ induces a map $\varphi_{(0)} : K^{r_1} \rightarrow K^{r_2}$, which is a map of vector spaces. Since localization is exact, φ is injective iff $\varphi_{(0)}$ is injective, which happens iff $r_1 \leq r_2$.

Exercise 7. (January 2017 Problem 5) Consider the ring $R = \mathbb{C}[x]$.

- (a) Describe all simple R -modules.
- (b) Give an example of an R -module that is indecomposable, but not simple. (Recall that a module is *indecomposable* if it cannot be written as a direct sum of non-trivial submodules.)
- (c) Consider R -modules $M = R/(x^3 + x^2)$, and $N = R/(x^3)$, and take their tensor product over R : $M \otimes_R N$. It is an R -module, and in particular a vector space over $\mathbb{C}$. What is its dimension over $\mathbb{C}$?

- (d) Let M be any R -module such that $\dim_{\mathbb{C}}(M) < \infty$, and let $N = R/(x^3)$ as before. Show that

$$\dim_{\mathbb{C}}(M \otimes_R N) = \dim_{\mathbb{C}} \operatorname{Hom}_R(N, M).$$

Solution 7.

- (a) Recall that a simple R -module is a non-zero R -module M whose only R -submodules are the 0 module and M itself. In this case, we know that a $\mathbb{C}[x]$ -module is the same thing as a complex vector space V together with a choice of a linear transformation $x: V \rightarrow V$. Suppose V is simple, then, since V is a complex vector space, we know that the linear transformation x has an eigenvector v of some eigenvalue λ . Then the subspace generated by v is stable under multiplication by x , and of course is stable under addition and scalar multiplication. So, $\langle v \rangle$ is a non-zero submodule of V , and so must in fact equal all of V . This shows that any simple R -module must be a 1-dimensional complex vector space. On the other hand, given any scalar $\lambda \in \mathbb{C}$, $\mathbb{C}$ considered as an R -module where $x: \mathbb{C} \rightarrow \mathbb{C}$ is multiplication by λ is a simple R -module, so we have described all the possible simple modules.
- (b) As we discovered in part (a), a submodule of an R -module V is a vector subspace which is stable under multiplication by x , and such subspaces are closely related to eigenspaces of x . Indeed, if x is diagonalizable, then its eigenvectors form a basis for V , which is the same as saying that V is the direct sum of the eigenspaces of x , which is to say that V is decomposable as an R -module. So, we certainly need x not to be diagonalizable, so consider $V = \mathbb{C}^2$ considered as an R -module with x acting as

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This module is not simple, because we already showed every simple module is 1-dimensional as a vector space. However, x has only one eigenspace, so V has only one R -submodule, and thus there's no way to write V as a direct sum of nontrivial R -submodules (we would need at least 2 submodules in order to do that). Thus, V is indecomposable.

- (c) Using standard properties of tensor products, we have

$$\begin{aligned} \mathbb{C}[x]/(x^3 + x^2) \otimes_{\mathbb{C}[x]} \mathbb{C}[x]/(x^3) &= \mathbb{C}[x]/(x^3, x^3 + x^2) \\ &= \mathbb{C}[x]/(x^2, x^3) \\ &= \mathbb{C}[x]/(x^2). \end{aligned}$$

Thus, $\dim_{\mathbb{C}}(M \otimes_R N) = \dim_{\mathbb{C}} \mathbb{C}[x]/(x^2) = 2$.

- (d) Here is the longer, but perhaps more “elementary” way to approach this one. Since $\dim_{\mathbb{C}}(M) < \infty$, in particular M must be finitely generated as an R -module. Since R is a PID, by the classification theorem, we must have that $M \cong \bigoplus_{i=1}^n (R/(f_i(x)))^{\oplus e_i}$ for some n and polynomials

$f_i(x)$. Then, as in part (c), we have

$$\begin{aligned} M \otimes_R N &\cong \left(\bigoplus_{i=1}^n \left(\frac{R}{(f_i(x))} \right)^{\oplus e_i} \right) \otimes_R N \\ &\cong \bigoplus_{i=1}^n \left(\frac{R}{(f_i(x))} \right)^{\oplus e_i} \otimes_R \frac{R}{(x^3)} \\ &\cong (R/(x))^{\oplus e_1} \oplus (R/(x^2))^{\oplus e_2} \oplus (R/(x^3))^{\oplus e_3}. \end{aligned}$$

Now, all you need to check really is that $\text{Hom}_R(N, R/(x^i))$ is an i -dimensional vector space for $i = 1, 2, 3$, and that the dimensions add correctly over these direct sums, and then you will conclude that $\dim_{\mathbb{C}}(M \otimes N) = e_1 + 2e_2 + 3e_3 = \dim_{\mathbb{C}} \text{Hom}_R(N, M)$.

Alternatively, we know that $M \otimes_R N = M \otimes_R (R/(x^3))$, and by a standard result, this is isomorphic to $M/(x^3)M$. On the other hand, to specify a function from $N = R/(x^3)$ to M , we just need to say where $1 \in R/(x^3)$ will map to, and the rest of the map will be determined by R -linearity. Furthermore, since $1 \in R/(x^3)$ satisfies $x^3 \cdot 1 = 0$, the image of 1 in M must be an element m satisfying $x^3 \cdot m = 0$. That is, $\text{Hom}_R(N, M) \cong M[x^3]$, the x^3 -torsion submodule of M . Notice that we have an exact sequence (but not a short exact sequence) of R -modules

$$0 \rightarrow M[x^3] \rightarrow M \xrightarrow{x^3} M \rightarrow M/(x^3)M \rightarrow 0.$$

In the case where $\dim_{\mathbb{C}} M < \infty$, this is also an exact sequence of finite-dimensional $\mathbb{C}$ -vector spaces. Since the sequence is exact, its Euler characteristic must be 0, so we have

$$\dim_{\mathbb{C}} M[x^3] - \dim_{\mathbb{C}} M + \dim_{\mathbb{C}} M - \dim_{\mathbb{C}} M/(x^3)M = 0.$$

Thus, $\dim_{\mathbb{C}} M[x^3] = \dim_{\mathbb{C}} M/(x^3)M$ as desired.

(Notice also that the module N is another example of an indecomposable R -module which is not simple.)

Exercise 8. (August 2017 Problem 1, edited) Let $M = \mathbb{C}[x]/(x^2 + x)$ and $N = \mathbb{C}[x]/(x - 1)$. Compute the dimension of the following tensor products as vector spaces over $\mathbb{C}$:

- (a) $M \otimes_{\mathbb{C}[x]} N$
- (b) $M \otimes_{\mathbb{C}} N$
- (c) $M \otimes_{\mathbb{C}[x^2]} N$.

Solution 8.

- (a) Recall that $R/I \otimes_R R/J \cong R/(I + J)$, so $M \otimes_{\mathbb{C}[x]} N \cong \mathbb{C}[x]/(x - 1, x^2 + x) = 0$. Thus, it is 0-dimensional over $\mathbb{C}$.
- (b) We have $M \otimes_{\mathbb{C}} N \cong M \otimes_{\mathbb{C}} \mathbb{C} \cong M$, which is 2-dimensional over $\mathbb{C}$ with basis $1, x$.

- (c) The only possible basis elements are $1 \otimes 1$ and $x \otimes 1$. Since we can multiply across the tensor by x^2 , we have $x \otimes 1 = -x^2 \otimes 1 = -(1 \otimes x^2) = -(1 \otimes 1)$. So this is 1-dimensional over $\mathbb{C}$ with basis $1 \otimes 1$.

Here is a more sophisticated way of doing this problem, which is not easier but which generalizes better to other situations involving tensor products. The ring $\mathbb{C}[x^2]$ is isomorphic to $\mathbb{C}[y]$. Under this identification, the inclusion $\mathbb{C}[x^2] \hookrightarrow \mathbb{C}[x]$ is the same as the composition of the inclusion $\mathbb{C}[y] \hookrightarrow \mathbb{C}[y][t]$ and the quotient $\mathbb{C}[y][t]/(t^2 - y)$. That is to say, we have a commuting square

$$\begin{array}{ccc} \mathbb{C}[y] & \hookrightarrow & \frac{\mathbb{C}[y][t]}{(t^2 - y)} \\ \downarrow \sim & & \downarrow \sim \\ \mathbb{C}[x^2] & \hookrightarrow & \mathbb{C}[x] \end{array} \quad \text{where the downward arrows are obtained by sending } y \mapsto x^2, t \mapsto x.$$

Now, we can rewrite the modules M and N as $\mathbb{C}[y]$ -modules:

$$M = \frac{\mathbb{C}[x]}{(x^2 + x)} \cong \frac{\mathbb{C}[y][t]}{(y + t, t^2 - y)} \cong \mathbb{C}[y], \quad N = \frac{\mathbb{C}[x]}{(x - 1)} \cong \frac{\mathbb{C}[y][t]}{(t - 1, t^2 - y)} \cong \frac{\mathbb{C}[y]}{(1 - y)}.$$

Thus, we get that $M \otimes_{\mathbb{C}[x^2]} N \cong M \otimes_{\mathbb{C}[y]} N \cong \mathbb{C}[y] \otimes_{\mathbb{C}[y]} \mathbb{C}[y]/(1 - y) \cong \mathbb{C}[y]/(1 - y)$, which is a 1-dimensional vector space.

Exercise 9. Let $f : M \times N \rightarrow P$ be an A -module homomorphism. For each $m \in M$, define a map $f_m : N \rightarrow P$ by $f_m(n) = f(m \otimes n)$. Show that this defines a isomorphism

$$\text{Hom}_A(M \otimes N, P) \cong \text{Hom}_A(M, \text{Hom}(N, P)).$$

Solution 9. For this to be an isomorphism, we need to define an inverse. So let ϕ be a homomorphism $M \rightarrow \text{Hom}_A(N, P)$ and let $b_\phi(x, y) = \phi(x)(y)$. Then b_ϕ is a bilinear map and therefore corresponds to some element of $\text{Hom}(M \otimes N, P)$ by the universal property.

Exercise 10. Show that if

$$M' \rightarrow M \rightarrow M'' \rightarrow 0$$

is exact, then so is

$$M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0.$$

Solution 10. Let f be the map $M' \rightarrow M$ and g be the map $M \rightarrow M''$. We want to check that $M'' \otimes N$ is the cokernel of the map $f \otimes 1$. Let $P = \text{im}(f \otimes 1)$. Now since $P = \text{im}(f \otimes 1) \subseteq \ker(g \otimes 1)$, the map $g \otimes 1 : M \otimes N \rightarrow M'' \otimes N$ induces a map $(M \otimes N)/P \rightarrow M'' \otimes N$. Let $h : M'' \otimes N \rightarrow (M \otimes N)/P$ be defined by sending a simple tensor $m'' \otimes n$ to $m \otimes n$ where $m \in M$ is such that $g(m) = m''$. We need to check that this doesn't depend on which m we choose; indeed, if m_1 and m_2 both have $g(m_1) = g(m_2) = m''$, then $g(m_1 - m_2) = 0$ so that $m_1 - m_2 \in \ker g = \text{im } f$ and $(m_1 - m_2) \otimes n \in P$ as required. Thus h is a well defined homomorphism, showing that $M'' \otimes N$ is indeed the cokernel of $f \otimes 1$.